

Option B: Clone via Reverse Engineering

Risk, scope, and comparison with the original-design track

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Companion to **CCLI-ROADMAP-001**

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Abstract

This document describes **Option B** for the CCI project: reverse engineering an existing commercial CCI-class product (AiLux, Selta, Energy Team, Higeco, or similar) with the intent to reproduce hardware layout, firmware behavior, or proprietary stack integration as closely as possible. It is a *separate* planning track from the primary roadmap (**CCLI-ROADMAP-001**), which targets an **original** SoM + carrier design driven by publicly documented CEI 0-16 functional requirements.

Option B may appear faster on paper but carries material intellectual-property, legal, certification, and long-term maintenance risk. Read this brief before committing budget or schedule to a clone path.

Not legal advice. This document is an engineering and project-planning summary only. Clone/reverse-engineering decisions require review by qualified IP and commercial counsel in your jurisdiction before purchase, teardown, or reproduction.

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1 Two project tracks at a glance

	Option A — Original design (CCLI-ROADMAP-001)	Option B — Clone / RE (this document)
Goal	CEI 0-16 compliant CCI from first principles; original schematic, carrier, firmware architecture	Reproduce a known commercial unit's HW/FW behavior with minimum functional deviation
Primary inputs	CEI Annex O/T, public interface specs, datasheets, your architecture choices	Purchased reference unit, manuals, teardown, protocol capture, binary/firmware analysis
Prototype cost	~€440–900 (single unit)	Reference unit purchase + RE tooling + likely respins when differences are found
Time to first demo	~10–12 weeks (balanced)	Can look faster if reference is stable; often slips once hidden dependencies surface
IP / legal posture	Lower — you own the design lineage	Higher — layout, BOM choices, firmware, and tooling may be protected
Certification story	Clean SDLC boundary; ATEX QMS/NB already in place; remaining scope is CEI/grid/cyber	Harder to prove independent design; lab scope may expand if behavior is “too similar”
Long-term margin	Amortize certification over your own BOM at volume	You may still pay commercial-grade BOM <i>and</i> carry RE/legal exposure

2 What “clone via reverse engineering” actually means

Option B is not the same as reading a user manual. In practice it usually spans some or all of the following workstreams.

2.1 Hardware reverse engineering

- **Non-destructive survey:** enclosure, connectors, labels, port count, isolation strategy, power input range.
- **Destructive teardown:** PCB photography, layer count, stackup inference, BOM extraction from markings.
- **Netlist / layout reconstruction:** redrawing schematic and PCB to match routing topology, impedance profiles, isolation barriers, and protection networks.
- **“Same look” carrier:** connector pinout, mechanical footprint, and silkscreen placement copied to ease field swap-in at MT plants.

Risk: PCB artwork, component placement for EMC/isolation, and some protection networks can be trade-secret or design-right protected even when the *function* (“4 Ethernet ports + RS-485 + DI/DO”) is industry-standard.

2.2 Firmware and software reverse engineering

- **Image extraction:** flash readout, eMMC dump, update-package capture, secure-boot bypass attempts.
- **Static/dynamic analysis:** disassembly, symbol recovery, configuration file formats, IEC 61850 logical node mapping.

- **Behavior cloning:** reproduce MMS data sets, GOOSE timing, PF2 curtailment state machine, Modbus maps, alarm semantics.
- **Stack replication:** if the reference uses a commercial IEC 61850 stack with vendor-specific integration, cloning behavior may effectively replicate proprietary glue code.

Risk: software copyright, license terms (GPL/commercial stack redistribution), and anti-circumvention rules around secure boot or encryption can all apply independently of whether the hardware “works the same.”

2.3 Integration and field-behavior cloning

- **DSO-side expectations:** some DSO test environments are tuned to specific vendor behaviors (reporting cadence, object naming, fault codes).
- **Commissioning tooling:** web UI flows, parameter sets, and project files may be vendor-specific.
- **Spare-part / service model:** a clone marketed as drop-in replacement can trigger unfair-competition or trademark issues even when CEI conformance is technically met.

3 Key risks (why this is the risky option)

3.1 Intellectual property and legal

Risk area	Typical concern
Copyright	Firmware, UI assets, documentation, and sometimes PCB artwork are protected expression.
Patents	Power-stage, isolation, or protocol-handling methods may be patented in EU/IT even if CEI function is standardized.
Trade secrets	BOM tuning, calibration data, and manufacturing know-how discovered by teardown are not automatically free to reuse.
Contract / license	Reference unit purchase, software updates, and protocol stacks may prohibit reverse engineering in EULA or supply terms.
Trademarking / passing off	Similar enclosure, labeling, or marketing as “compatible replacement” can create separate commercial-law exposure.
Export / security law	Breaking secure boot or extracting keys may intersect with cybersecurity regulations beyond product compliance.

3.2 Certification and audit

- **IEC 62443-4-1** audits your *development process*. A clone program built from stolen or unlicensed reference artifacts is difficult to document cleanly.
- **IEC 62443-4-2 / IEC 62351** scope grows if you cannot explain security architecture independently of the reference product.
- **IEC 61850 conformance** proves protocol behavior, not ownership. Passing a conformance suite does not cure IP problems.
- **DSO acceptance** is partly political: utilities may reject a device that appears to be an unauthorized copy even if lab tests pass.

3.3 Engineering and business

- **Hidden dependencies:** calibration, factory scripts, cloud keys, or licensed modules may not be visible until late in replication.
- **Update fragility:** when the reference vendor ships a firmware revision, your clone can break in the field unless you re-RE each time.
- **No cost advantage at low volume:** commercial CCIs are expensive because certification and liability are bundled in — cloning does not remove certification cost; it often *adds* RE cost on top.
- **Support burden:** you inherit every undocumented edge case the original vendor already learned over years of deployments.

4 What is usually safe to reuse (even on Option B)

Generally lower-risk inputs: public standards (CEI 0-16, IEC 61850, IEC 62351, IEC 62443), datasheet-level component choices, publicly documented interface requirements, and *functional* observations (port count, voltage class, protocol types) without copying expression of the implementation.

- CEI 0-16 Annex O/T/M text and publicly published FAQ interpretations.
- Open-source stacks used under their license (e.g. `libiec61850` with MPL obligations respected).
- Commodity blocks (SoM, PHY, RS-485 transceiver, GNSS module) selected from public datasheets.
- Black-box interface testing: observe *what* appears on Eth_A/Eth_B/RS-485 without copying internal code.

This overlap is why **Option A** can reach the same field behavior without Option B's legal surface area.

5 Option B workflow (if you still pursue it)

1. **Legal gate 0:** written IP review; confirm purchase/EULA terms on the reference unit and any software images.
2. **Reference acquisition:** buy at least one production-revision unit; log serial, firmware version, and options.
3. **Non-destructive characterization:** photos, port map, power draw, boot logs, network capture on a isolated bench VLAN.
4. **Controlled teardown:** document layers and BOM; do *not* assume 1:1 layout reproduction is permitted.
5. **Functional matrix:** spreadsheet mapping every CEI 0-16 requirement to observed reference behavior vs. your planned implementation.
6. **Clean-room split (strongly recommended):** one team/person set exposed to reference internals; a separate team implements from specs only.
7. **Original artifacts only in deliverables:** new schematic sheets, new PCB, new source tree, new ICD/CID files — not traced copies.
8. **Certification planning in parallel:** scope 62443/61850/62351 assuming you must defend independent design.

6 Rough effort and cost (Option B only)

Workstream	Indicative effort	Notes
Reference hardware + spares	€ 2,000–8,000+	Depends on vendor, lead time, and whether MT/POC configuration is required
RE tooling (scope, logic analyzer, JTAG/flash tools)	€ 500–3,000	Often already owned in a HW lab
HW teardown + documentation	2–6 weeks	Photography, BOM, interface map
Schematic/PCB “similarity” layout	4–10 weeks	High IP risk if copied; faster but legally dangerous
FW/protocol capture + behavior model	6–16 weeks	IEC 61850 + PF2 logic is non-trivial
Clean re-implementation	8–20 weeks	Parallel to above if clean-room enforced
Certification (same as Option A)	€ 50,000–120,000	Not avoided by cloning; may increase if independence cannot be shown
Legal / IP counsel	€ 5,000–25,000+	Should be budgeted up front, not after a cease-and-desist
Realistic incremental cost vs Option A	€ 10,000–40,000+ before certification	Plus schedule risk and ongoing re-RE tax on vendor updates

7 Decision guide

Choose Option A (original design) if you want:

- a defensible product line you can certify, sell, and maintain independently;
- lower IP exposure and a clearer IEC 62443-4-1 story;
- freedom to optimize BOM for your target cost at volume.

Option B only makes sense as a time-boxed investigation if:

- you legally own or have licensed the reference artifacts;
- a customer explicitly requires bit-compatible behavior and accepts the liability;
- you use RE output strictly to populate a *functional requirements matrix*, then discard traceable copies before implementation;
- counsel signs off on clean-room boundaries and go-to-market positioning (not “fake AiLux”).

Recommended project default: treat Option B as a *restricted research spike* (4–6 weeks max) producing a requirements and risk report — not as the main product architecture. Implement on Option A unless legal and commercial gates explicitly approve otherwise.

8 Suggested deliverables if Option B is authorized

1. **CCLI-RE-IF-001** — Interface & port survey (non-destructive).
2. **CCLI-RE-BOM-001** — Component class list (*functional equivalents*, not 1:1 part numbers copied from silkscreen unless public datasheet parts).
3. **CCLI-RE-PROT-001** — Protocol capture report (MMS/GOOSE/Modbus/PF2 observations).
4. **CCLI-RE-RISK-001** — IP/legal red-flag memo from counsel.
5. **Handoff to Option A** — update **CCLI-ROADMAP-001** pin budget and firmware boundary; *no* traced schematic imports.

9 Related documents

- **CCLI-ROADMAP-001** — CCI_Project_Roadmap.tex / .pdf (primary track: original design)
- **CCI_Architecture.mermaid** — functional block diagram (English)
- Telemetry RAG sources: `cei-0-16-allegato-o`, `cei-0-16-allegato-t`, `cei-0-16-cci-manual-ailux`